

Math 54 Discussion Worksheet

Name: _____

Problem 1

- (a) Suppose the second column of B is all zeros. What can you say about the second column of AB ?
- (b) Suppose the last column of AB is all zeros, but B itself has no column of zeros. What can you say about the columns of A ?

Problem 2

Show that if AB is invertible, then so are both A and B .

Problem 3

Suppose A , B , and X are $n \times n$ matrices with A , X , and $A - AX$ invertible, and suppose

$$(A - AX)^{-1} = X^{-1}B. \tag{1}$$

- (a) Explain why B is invertible.
- (b) Solve (1) for X . If you need to invert a matrix, explain why that matrix is invertible.